

PROPOSAL PRESENTATION

ON

DEVELOPMENT OF A LOG ANALYSIS SYSTEM FOR
DETECTING SUSPICIOUS ACTIVITIES FROM LINUX
SERVERS AND WEB BASED LOGS

by

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AIM AND OBJECTIVES

Aim

Development of a log analysis System for detecting suspicious activities from linux servers and web based logs

The objectives are to:

- i. develop a system that collects, preprocesses, and parses log data automatically.
- ii. implement Drain3 log parsing for real-time generation of structured log templates.
- iii. design a rule-based system to detect known suspicious activities, like failed logins and unauthorized access, alongside an Isolation Forest model to flag unknown anomalies.

SCOPE OF THE STUDY

- System Scope: Focused on analyzing Linux server logs (auth logs, syslogs) and web server logs (Apache/Nginx access and error logs) for suspicious activities.
- Data Scope: Uses real log files collected from Linux-based environments. Log formats covered include: /var/log/auth.log, /var/log/syslog, and standard web server log formats.
- Conceptual Scope: Centers on identifying suspicious activities such as failed logins, privilege escalation attempts, repeated errors, unusual traffic patterns, and other anomaly indicators.
- Methodological Scope: Implements a lightweight pipeline combining log parsing(using drain3), rule-based rules(rule-based detection), and machine-learning-based anomaly detection(using isolation forest model). Evaluation will be performed using 300–500 log samples.

PROPOSED METHODOLOGY

❖ Log Collection

This is the stage where the real server logs such as Linux authentication logs, syslogs, and web server logs, and store all the collected logs in a well-structured directory to make proper analysis easier and more organized.

❖ Preprocessing

Preprocessing would involve the cleaning of the logs and preparing them for parsing by removing the dynamic parts(logs that keep changing no specific value), normalizing the text, standardizing the timestamp formats, and performing tokenization to ensure the data is consistent and ready for further analysis.

METHODOLOGY CONT'D

❖ Log Parsing using Drain3

This step involves using the Drain3 algorithm to parse the logs by converting them into templates, grouping similar log messages into meaningful patterns, and producing structured logs that are stored as JSON to make further analysis easier and more organized.

❖ Detection Algorithms

The step covers the detection algorithms, starting with rule-based detection, where you create simple rules to catch suspicious activities such as more than five failed logins in one minute, users trying to access root privileges, or multiple repeated errors coming from the same IP address.

It also includes using an Isolation Forest model, which is trained on normal logs to identify unusual behavior by isolating outliers and generating an anomaly score for each log entry.

PROPOSED SYSTEM ARCHITECTURE

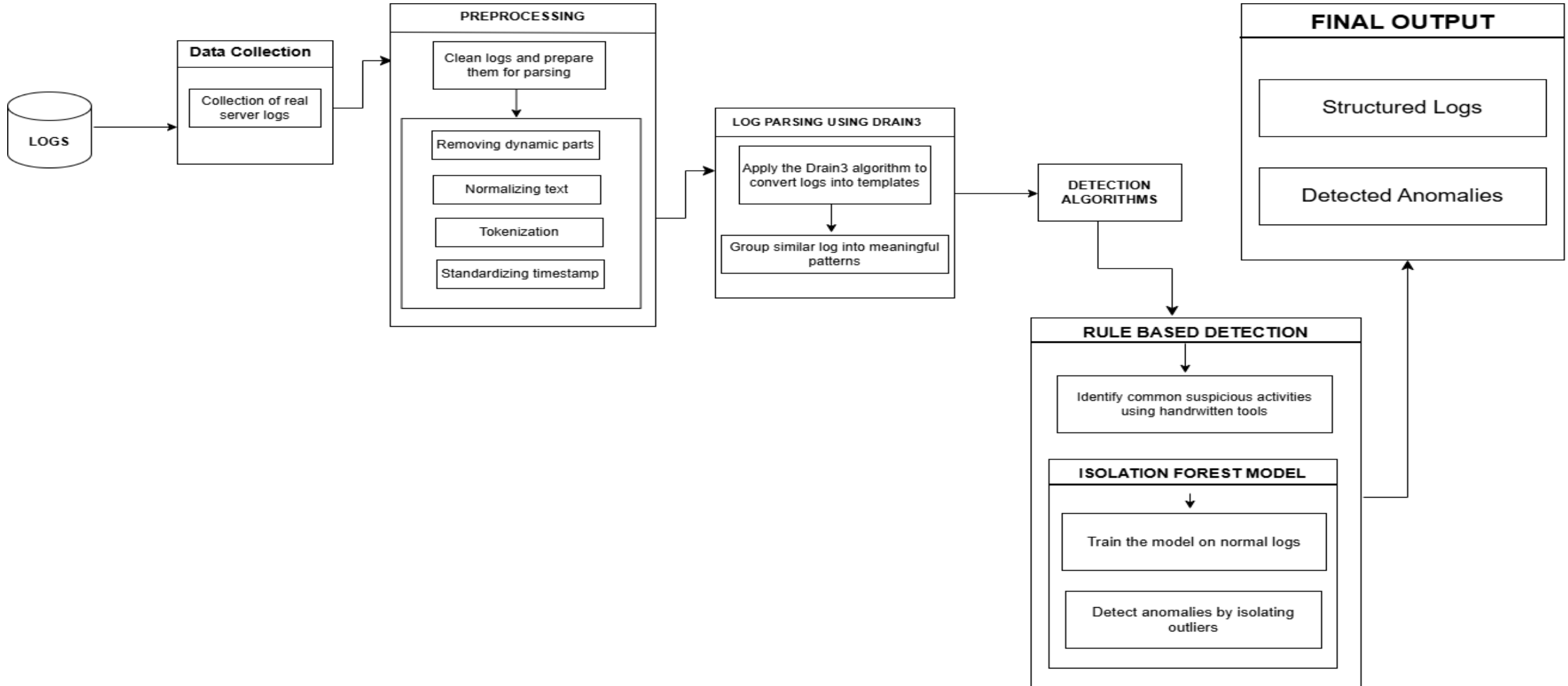


Figure 1: Diagram of The Proposed System Architecture